NexFinance Module Guide

Dokumen ini adalah panduan operasional untuk seluruh modul NexFinance (frontend + backend API).

1. Ringkasan Sistem

- **Frontend**: React SPA
- **Backend**: Express REST API
- **Base API**: `/api/v1`
- **Auth**: JWT Bearer Token
- **Authorization**: RBAC per modul
- **Audit**: semua aksi penting dicatat ke Activity Logs

2. Matriks Modul

Modul	Tujuan	Route Frontend	Endpoint Utama
Authentication	Login, register, verifikasi token	`/login`	`/auth/*`
Dashboard	KPI dan ringkasan finansial	`/dashboard`	`/dashboard/kpi`, `/dashboard/monthly-report`
SPP	Invoice, pembayaran, bukti bayar	`/spp`	`/spp/invoices`, `/spp/payment`, `/spp/reports`
Payroll	Hitung gaji dan histori payroll	`/payroll`	`/payroll/employees`, `/payroll/calculate`, `/payroll/history`, `/payroll/reports`
Expenses	Pengajuan dan approval pengeluaran	`/expenses`	`/expenses`, `/expenses/submit`, `/expenses/:id/approve`, `/expenses/:id/reject`
Procurement	PR, PO, penerimaan barang	`/procurement`	`/procurement/pr`, `/procurement/po`, `/procurement/po/:id/receive`, `/procurement/reports`
Accounting	COA, jurnal, laporan akuntansi	`/accounting`	`/accounting/chart-of-accounts`, `/accounting/journals`, `/accounting/reports/*`
Reports	Laporan finansial lintas modul	`/reports`	konsumsi endpoint report di modul terkait
Master Data	Siswa, karyawan, user, role matrix	`/master/students`, `/master/employees`, `/master/access`	`/master/students`, `/master/employees`, `/master/users`, `/master/role-permissions/*`

Activity Logs	Audit trail & statistik aktivitas	`/activity-logs`	`/activity`, `/activity/stats`, `/activity/clear`
Notifications	Konfigurasi dan kirim notif	`/notifications`	`/notifications/config`, `/notifications/send`

3. Detail Per Modul

3.1 Authentication

Tujuan

- Autentikasi user dan penyediaan data role + permissions.

Endpoint

- `POST /api/v1/auth/login`
- `POST /api/v1/auth/register`
- `GET /api/v1/auth/verify`
- `GET /api/v1/auth/me`
- `GET /api/v1/auth/permissions`

Catatan

- `login/register` menghasilkan JWT.
- `me/permissions` mengembalikan permission modul berdasar role aktif.

3.2 Dashboard

Tujuan

- Menampilkan KPI utama (pemasukan, collection rate, pending items, dsb).

Endpoint

- `GET /api/v1/dashboard/kpi`
- `GET /api/v1/dashboard/monthly-report`

Output kunci

- `monthRevenue`, `collectionRate`, `pendingItems`, `breakdown`.

3.3 SPP

Tujuan

- Kelola invoice SPP, pembayaran, dan bukti bayar.

Endpoint

- `GET /api/v1/spp/invoices`
- `POST /api/v1/spp/invoices`
- `POST /api/v1/spp/payment`
- `GET /api/v1/spp/reports`

Flow inti

1. Buat invoice.
2. Rekam pembayaran (`/payment`).
3. Jika `notify=true` , sistem mengirim notifikasi email/WA.
4. Backend mengembalikan payload `receipt` untuk cetak bukti.

3.4 Payroll

Tujuan

- Hitung payroll karyawan per periode dan simpan histori.

Endpoint

- `GET /api/v1/payroll/employees`
- `POST /api/v1/payroll/calculate`
- `GET /api/v1/payroll/history`
- `GET /api/v1/payroll/reports`

Catatan

- `calculate` membutuhkan `employeeId` dan `month` .
- Hasil tersimpan sebagai payroll run.

3.5 Expenses

Tujuan

- Pengajuan biaya operasional dan approval bertingkat.

Endpoint

- `GET /api/v1/expenses`
- `POST /api/v1/expenses/submit`
- `POST /api/v1/expenses/:id/approve`
- `POST /api/v1/expenses/:id/reject`

Catatan

- Expense status: `pending` , `approved` , `rejected` .
- Approval count menentukan final status approved.

3.6 Procurement

Tujuan

- Kelola pengadaan dari Purchase Request ke Purchase Order.

Endpoint

- `GET /api/v1/procurement/pr`
- `POST /api/v1/procurement/pr`
- `GET /api/v1/procurement/po`
- `POST /api/v1/procurement/po`
- `POST /api/v1/procurement/po/:id/receive`
- `GET /api/v1/procurement/reports`

Flow inti

1. Buat PR.
2. Buat PO dari PR.
3. Tandai PO diterima.

3.7 Accounting

Tujuan

- Pencatatan akuntansi dan output laporan keuangan.

Endpoint

- `GET /api/v1/accounting/chart-of-accounts`
- `GET /api/v1/accounting/journals`
- `POST /api/v1/accounting/journals`
- `GET /api/v1/accounting/reports/balance-sheet`
- `GET /api/v1/accounting/reports/income-statement`

3.8 Reports

Tujuan

- Menyediakan dashboard laporan lintas modul.

Sumber data

- SPP reports
- Payroll reports
- Procurement reports
- Accounting reports
- Dashboard monthly report

3.9 Master Data

Tujuan

- Menjadi pusat data referensi (siswa, karyawan, user, role permissions).

Sub modul

- **Master Siswa**: CRUD + filter/search.
- **Master Karyawan**: CRUD data karyawan.
- **Hak Akses Role**:
  - kelola role user (khusus super admin untuk edit role user)
  - kelola matriks akses role → module (khusus super admin)

Endpoint

- `GET/POST/PUT/DELETE /api/v1/master/students`
- `GET/POST/PUT/DELETE /api/v1/master/employees`
- `GET/POST/PUT/PATCH /api/v1/master/users*`
- `GET /api/v1/master/role-permissions`
- `PATCH /api/v1/master/role-permissions/:role`

3.10 Activity Logs

Tujuan

- Menyimpan jejak aktivitas untuk audit dan troubleshooting.

Endpoint

- `GET /api/v1/activity`
- `GET /api/v1/activity/stats`
- `DELETE /api/v1/activity/clear` (admin)

Filter

- `module`, `action`, `search`, `limit`, `offset`

3.11 Notifications

Tujuan

- Kirim notifikasi manual/otomatis via Email dan WhatsApp.

Endpoint

- `GET /api/v1/notifications/config`
- `POST /api/v1/notifications/send`

Payload minimum

- `channels` (`email`, `whatsapp`)
- `message`
- `to.email` dan/atau `to.phone`

4. RBAC dan Hak Akses

Role default:

- `super_admin`, `admin`, `bendahara`, `kabag`, `staff`, `auditor`

Permission map utama berada di:

- `backend/src/middleware/rbac.js`

Matriks akses bersifat dinamis melalui endpoint:

- `PATCH /api/v1/master/role-permissions/:role`

5. Integrasi Notifikasi (Email & WhatsApp)

Konfigurasi di `backend/.env`:

- Email:
  - `EMAIL_PROVIDER_URL` atau `EMAIL_WEBHOOK_URL`
  - `EMAIL_PROVIDER_TOKEN` (opsional sesuai provider)
- WhatsApp:
  - `WHATSAPP_META_TOKEN`
  - `WHATSAPP_PHONE_NUMBER_ID`

- `WHATSAPP_WEBHOOK_URL` (fallback opsional)

6. Standar Operasional Harian (Saran)

1. Login dengan role sesuai tugas.
2. Monitor KPI di Dashboard.
3. Proses transaksi harian (SPP/Payroll/Expenses/Procurement).
4. Review laporan di Reports/Accounting.
5. Cek Activity Logs untuk audit.
6. Gunakan Notifications untuk reminder atau bukti kirim info.

7. Troubleshooting Cepat

- **401 Unauthorized**: cek token login / masa berlaku JWT.
- **403 Access denied**: role tidak memiliki akses modul.
- **Notifikasi gagal**: validasi env provider dan endpoint webhook.
- **Data tidak muncul**: cek filter aktif, role permission, dan log activity.

8. Referensi File Kode

- Backend routes: `backend/src/routes/*.js`
- RBAC middleware: `backend/src/middleware/rbac.js`
- Frontend routes: `frontend/src/App.js`
- Service API frontend: `frontend/src/services/api.js`

Dokumen ini dapat dijadikan SOP teknis awal untuk onboarding, operasional harian, dan audit internal modul NexFinance.